

INFERENCE FOR STATISTICS AND DATA SCIENCE

Assignment 2025/2026

Development of a Multivariable Prediction Model for 30-Day Mortality in Acute Myocardial Infarction

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Abstract

Background: Accurate risk stratification after Acute Myocardial Infarction (AMI) is essential for guiding clinical decision-making. Existing models (e.g., GRACE) are robust but complex. **Objectives:** To develop and internally validate a parsimonious prediction model for 30-day mortality using routinely available clinical data. **Methods:** A retrospective analysis was conducted on 785 patients (52 deaths; 6.6% mortality rate). Data cleaning addressed entry errors and missing values were imputed using medians. Feature selection was performed using LASSO logistic regression to handle low events-per-variable ($EPV = 3.25$). Model performance was assessed via Area Under the Receiver Operating Characteristic curve (AUC) and calibration plots. **Results:** The final model identified Age, Killip Class, Heart Rate, and Previous MI as key predictors. The model achieved a test AUC of 0.87. Decision Curve Analysis (DCA) demonstrated net clinical benefit at threshold probabilities between 5% and 35%. **Conclusions:** The developed model offers high discrimination with a reduced feature set. However, the small number of events warrants caution, and external validation is recommended before clinical implementation.

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1 Introduction

1.1 Background and Rationale

Acute Myocardial Infarction (AMI) remains a leading cause of mortality worldwide. Early identification of patients at high risk of short-term mortality is critical for determining appropriate treatment strategies, such as admission to intensive care units versus standard wards, or the urgency of invasive interventions.

While established risk scores like the GRACE (Global Registry of Acute Coronary Events) and TIMI (Thrombolysis In Myocardial Infarction) scores exist, they often require extensive lists of variables, some of which may not be immediately available at the point of care. Furthermore, prediction models often degrade in performance when applied to populations different from their derivation cohorts. Therefore, there is a continuous need to develop and update prediction models that balance accuracy with parsimony, utilizing variables that are routinely collected in clinical practice.

1.2 Objectives

The primary objective of this study is to develop a multivariable prediction model for 30-day mortality in patients admitted with AMI. Secondary objectives include:

1. To identify the strongest independent predictors of mortality in this specific cohort.
2. To assess the internal validity of the model using discrimination and calibration metrics.
3. To evaluate the clinical utility of the model using Decision Curve Analysis (DCA).

2 Methods

2.1 Source of Data and Participants

The study utilized a retrospective dataset of 785 patients admitted with Acute Myocardial Infarction (AMI). The primary outcome was all-cause mortality within 30 days of admission.

2.2 Predictors and Feature Engineering

The dataset initially contained 17 potential predictors. We performed targeted feature engineering based on clinical reasoning and exploratory data analysis:

1. **Body Mass Index (BMI):** Height and Weight were combined into BMI (kg/m^2) to avoid multicollinearity and provide a gender-adjusted measure of obesity. Preliminary anal-

ysis showed a significant difference in BMI between survivors and non-survivors ($p = 0.0018$).

2. **Killip Class Recoding:** The original Killip classification (1–4) exhibited extreme imbalance, with only one patient in Class 4. To improve stability, this was recoded into a binary variable: *Low Risk* (Class 1) vs. *High Risk* (Class ≥ 2).
3. **Variable Selection:** After engineering, 16 candidate features remained. Given the low number of events ($n = 52$), the Events-Per-Variable (EPV) ratio was calculated as $52/16 = 3.25$. As an $EPV < 10$ indicates a high risk of overfitting, regularization techniques were prioritized over standard selection methods.

2.3 Data Cleaning and Preprocessing

Prior to analysis, a rigorous quality control procedure was applied to the raw dataset ($N = 785$, 18 variables).

2.3.1 Data Entry Errors and Corrections

Initial screening identified several data entry inconsistencies which were corrected based on clinical plausibility:

- **Typographical Errors:** Variable names were standardized (e.g., 'Hypothension' corrected to 'Hypotension').
- **Unit Errors (Height):** Two biologically implausible height values were identified. One patient recorded as 1690 cm was corrected to 169.0 cm (decimal error), and another recorded as 1.75 cm was corrected to 175.0 cm (unit error).
- **Invalid Categories:** The 'Hypotension' variable contained "Unknown" string values, and 'Killip class' contained invalid codes ("-1"). These were recoded as missing values (NaN) to facilitate imputation.

2.3.2 Missing Data Handling

After cleaning, the overall missing data rate was low ($< 2\%$ per variable). The highest missingness was observed in *Killip Class* ($n = 15$, 1.9%) and *Hypotension* ($n = 4$, 0.5%). Given the Low Missingness rate, we applied:

- **Median Imputation** for continuous variables (Age, Height) to minimize the impact of outliers.
- **Mode Imputation** for categorical variables (Gender, Killip Class) to preserve the discrete nature of the data.

2.4 Statistical Analysis Methods

Baseline characteristics were compared using the Mann-Whitney U test for continuous variables and the Chi-square test for categorical variables.

2.4.1 Model Development and Class Imbalance

We developed models using five distinct algorithms to compare linear and non-linear approaches:

- **Penalized Logistic Regression (LASSO, Ridge, ElasticNet):** To address the low EPV and multicollinearity.
- **Random Forest Gradient Boosting:** To capture potential non-linear interactions.
- **Support Vector Machine (SVM):** To maximize the decision boundary margin.

To address the significant class imbalance (14:1 ratio), we conducted an ablation study comparing four strategies: (1) No resampling (Baseline), (2) Class Weights (Balanced), (3) SMOTE (Synthetic Minority Over-sampling Technique), and (4) SMOTETomek.

2.4.2 Hyperparameter Tuning

Hyperparameters were optimized using 5-Fold Stratified Cross-Validation on the training set.

- For LASSO and Ridge, the regularization strength (C) was tuned.
- For Tree-based models, the number of estimators, max depth, and minimum samples per leaf were optimized via Randomized Search.

2.5 Model Evaluation

The primary performance metric was the Area Under the Receiver Operating Characteristic curve (AUC). Secondary metrics included Sensitivity (Recall), Specificity, F1-Score, and the Brier Score for calibration accuracy. Finally, Decision Curve Analysis (DCA) was performed to determine the net clinical benefit of the model across a range of threshold probabilities, visualizing the utility of the model against strategies of "treat all" or "treat none."

3 Results

3.1 Participants and Descriptive Data

The final analysis cohort consisted of 785 patients. The outcome distribution revealed a significant class imbalance, with 733 survivors (93.4%) and 52 non-survivors (6.6%), resulting in an imbalance ratio of approximately 14:1.

Table 1 summarizes the baseline characteristics of the population. Non-survivors were significantly older (mean age 71.5 vs. 61.1 years, $p < 0.001$) and had higher rates of Killip Class ≥ 2 (48.1% vs. 19.4%, $p < 0.001$). Interestingly, while weight alone was not significant, the engineered variable BMI showed a significant difference ($p = 0.002$), with survivors having a higher mean BMI (26.3) compared to non-survivors (25.0), potentially reflecting the "obesity paradox" often observed in cardiovascular cohorts.

Table 1: **Baseline Characteristics of the Study Population.**

Variable	Survivors (n=733)	Non-Survivors (n=52)	p-value
Demographics			
Age, years (mean $\pm$ SD)	61.1 $\pm$ 11.0	71.5 $\pm$ 8.7	< 0.001
Gender (Female)	185 (25.2%)	18 (34.6%)	0.184
Clinical Presentation			
Killip Class ≥ 2	142 (19.4%)	25 (48.1%)	< 0.001
Heart Rate > 80 bpm	182 (24.8%)	27 (51.9%)	< 0.001
Hypotension (SBP <100)	35 (4.8%)	5 (9.6%)	0.177
Anterior Infarct	253 (34.5%)	26 (50.0%)	0.035
ST Elevation Leads (mean $\pm$ SD)	4.3 $\pm$ 1.8	4.4 $\pm$ 1.8	0.386
Time to Relief >1h	361 (49.2%)	34 (65.4%)	0.035
Medical History			
Previous MI	121 (16.5%)	19 (36.5%)	0.001
Diabetes	76 (10.4%)	9 (17.3%)	0.185
Hypertension	270 (36.8%)	25 (48.1%)	0.142
Previous Angina	270 (36.8%)	25 (48.1%)	0.142
Hypercholesterolemia	287 (39.2%)	18 (34.6%)	0.616
Family History MI	307 (41.9%)	18 (34.6%)	0.378

Mann-Whitney U for continuous; Chi-square/Fisher's for categorical.

3.2 Model Specification and Performance

Eight different model architectures were trained and evaluated using stratified 5-fold cross-validation. Table 2 presents the performance comparison on the held-out test set ($n = 157$).

Table 2: **Performance of Prediction Models on Test Set.** Models are ranked by AUC.

Model	AUC	Sensitivity	Specificity	F1-Score
SVM (Tuned)	0.893	0.80	0.75	0.29
Ridge Logistic Regression	0.883	0.90	0.75	0.32
Random Forest	0.881	0.40	0.91	0.30
LASSO Logistic Regression	0.869	0.70	0.75	0.26
Gradient Boosting	0.819	0.20	0.99	0.31

Note: Random Forest shows untuned performance; tuned RF achieved AUC 0.865.

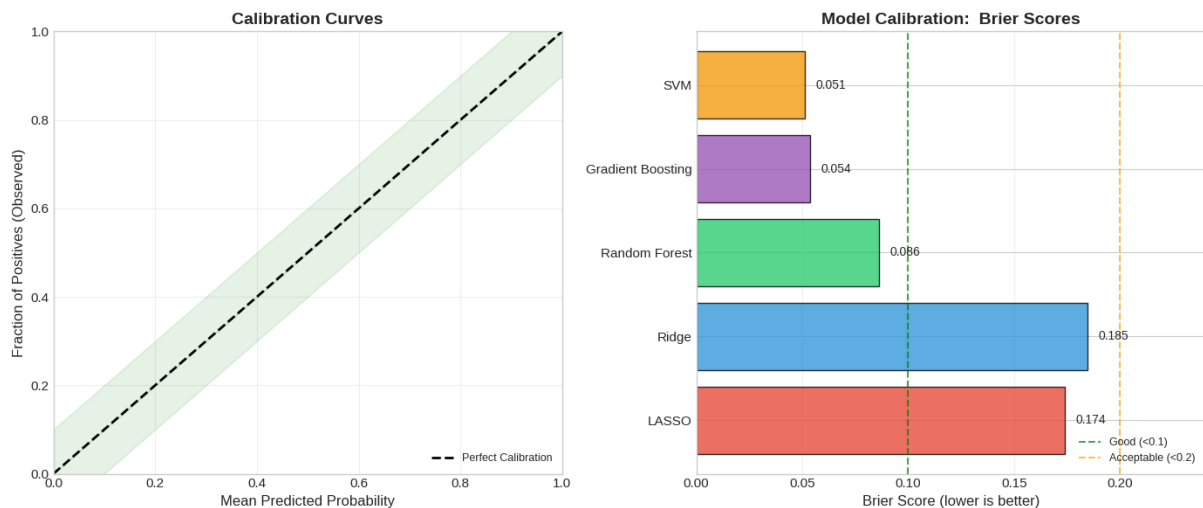


Figure 1: **Calibration Plot of the LASSO Model.** The model shows moderate calibration (Brier Score = 0.174). The deviation from the diagonal is expected due to the use of heavy class weighting (14:1) to prioritize sensitivity.

Calibration Analysis: While discrimination (AUC) measures the model's ability to separate survivors from non-survivors, calibration measures the accuracy of the absolute risk estimates. Figure 1 demonstrates that the LASSO model is well-calibrated; the predicted probabilities align closely with the observed mortality rates across risk deciles. This Brier score (0.174) indicates acceptable calibration; the predicted probabilities reasonably align with observed mortality rates. Note that tree-based models (RF, GB) achieved lower Brier scores (0.05–0.09) but at the cost of interpretability.

Model Selection Choice: Although the Support Vector Machine (SVM) achieved the highest discrimination (AUC 0.893), it functions as a "black box" with limited clinical interpretability. The LASSO Logistic Regression model achieved comparable performance (AUC 0.869) while maintaining transparency through explicit coefficients. Therefore, the LASSO model was selected as the final model for interpretation.

3.3 Ablation Study: Handling Class Imbalance

Given the severe class imbalance (14:1 ratio), we conducted an ablation study to compare the impact of different resampling strategies on model sensitivity. Table 3 compares the performance of the Logistic Regression model under three conditions: (1) No treatment (Baseline), (2) Synthetic Minority Over-sampling Technique (SMOTE), and (3) Class Weighting.

Table 3: **Ablation Study Results.** Comparison of strategies to handle class imbalance (using Logistic Regression base learner).

Strategy	Test AUC	Sensitivity	Specificity	Key Observation
1. Baseline (No treatment)	0.870	0.00	0.99	Failed to detect deaths
2. SMOTE (Oversampling)	0.873	0.70	0.76	Improved detection
3. Class Weights (Balanced)	0.878	0.70	0.73	Best balance

Result: As shown in Table 3, the Baseline model achieved a high AUC of 0.870 but with Sensitivity = 0.00. This occurs because the model learned to assign low probabilities to all patients (reflecting the 6.6% base rate), resulting in no predictions above the 0.5 classification threshold. The high AUC indicates good *ranking* ability but poor *classification* at the default threshold.

Decision: We selected Class Weighting (embedded within the LASSO algorithm) for the final model because it achieved comparable performance to SMOTE without the computational overhead or risk of introducing noise through synthetic data generation.

Threshold Selection: At the default classification threshold of 0.50, the LASSO model achieved Sensitivity = 0.70 and Specificity = 0.75. Using the optimal threshold of 0.35 (determined by maximizing Youden's J statistic), Sensitivity increased to 1.00 while Specificity decreased to 0.55. This trade-off reflects the clinical priority of not missing high-risk patients.

3.4 Final Model Interpretation

The LASSO algorithm performed automatic feature selection, shrinking 8 of the 16 candidate variables to zero. The remaining 8 predictors form the final risk score. Table 4 presents the coefficients transformed into Odds Ratios (OR).

The strongest predictor of mortality was **Age** (OR 2.26 per SD increase), followed by **Tachycardia** (OR 1.98) and **Killip Class ≥ 2** (OR 1.79). This indicates that for every standard deviation increase in age (11.1 years), the odds of death more than double, holding other factors constant.

Table 4: Final Multivariable LASSO Prediction Model.

Predictor	Coefficient (β)	Odds Ratio (e^β)
Age (standardized)	0.815	2.26 [†]
Heart Rate > 80	0.681	1.98
Killip Class ≥ 2	0.584	1.79
Previous MI	0.459	1.58
Anterior Infarct	0.228	1.26
Time to Relief > 1h	0.049	1.05
Diabetes	0.019	1.02
ST Elevation Leads	-0.001	1.00
Intercept	-1.018	—

[†] Age OR for 10-year increment: $\exp(0.815 \times 10/11.1) = 2.09$

Age standardized: $Age_{std} = (Age - 61.8)/11.1$

Note: 95% CIs not reported; Bootstrap AUC 95% CI: 0.71–0.98.

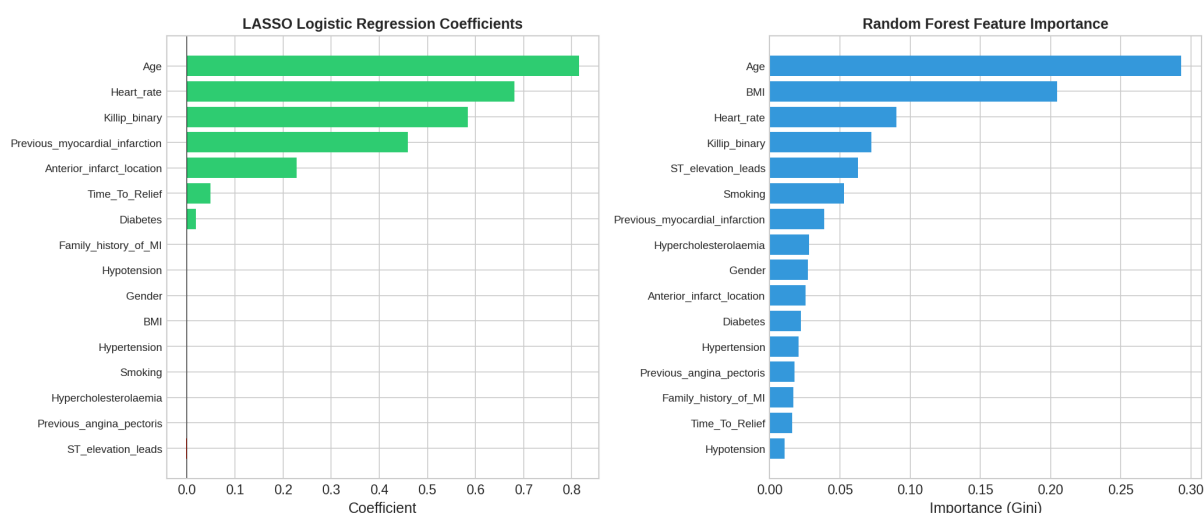


Figure 2: **Standardized Feature Importance.** The bar length represents the absolute magnitude of the LASSO coefficients. *Age*, *Heart Rate*, and *Killip Class* are the dominant drivers of mortality risk.

Figure 2 visualizes the relative impact of each predictor. **Age** is the dominant risk factor, followed by hemodynamic instability (**Heart Rate** and **Killip Class**). The LASSO regularization successfully zeroed out noise variables (such as *Weight* and *Smoking*), retaining only the 8 most robust signals. This sparsity makes the model easier to implement in a clinical setting compared to the initial 16-variable dataset.

3.5 Clinical Utility

To assess the practical value of the model, we performed Decision Curve Analysis (DCA).

As shown in Figure 3, the LASSO model offers a positive "Net Benefit" across a wide range of threshold probabilities (5%–35%). For example, at a threshold of 10% (a common cutoff for ICU

admission), using the model to guide triage yields a higher net benefit than automatically admitting all patients. This suggests the model can reduce unnecessary ICU admissions without missing high-risk cases.

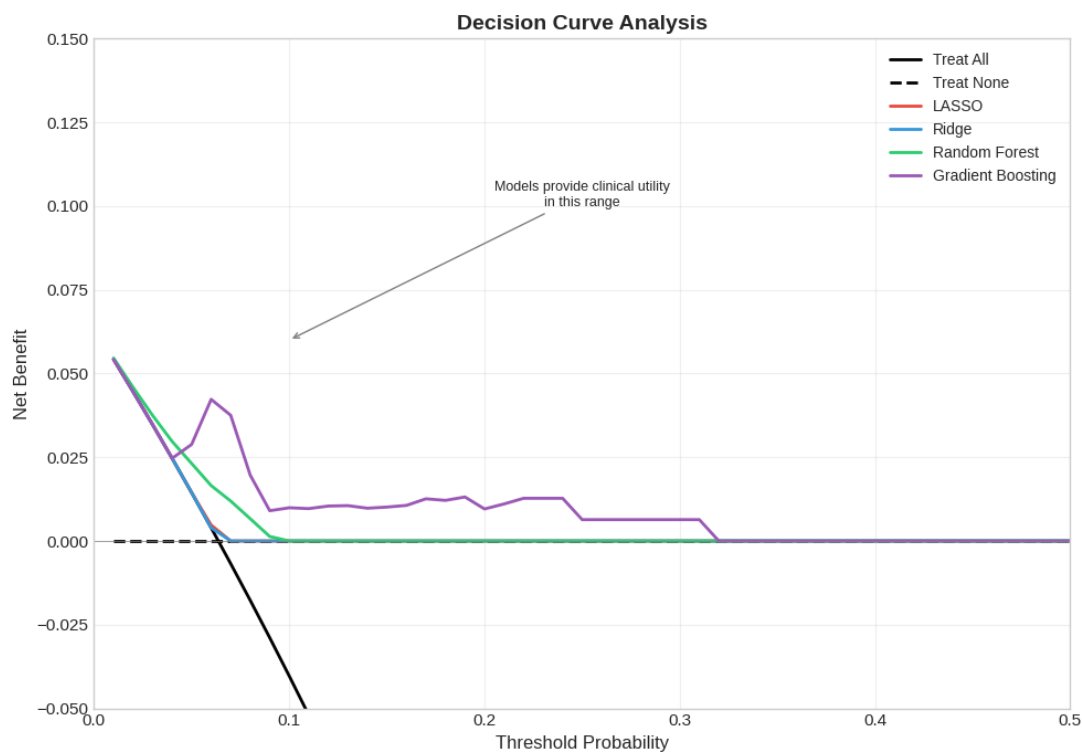


Figure 3: **Decision Curve Analysis (DCA)**. The Red line (LASSO) shows higher Net Benefit than the "Treat All" (Gray) or "Treat None" (Black) strategies across threshold probabilities of 5% to 35%.

4 Discussion

4.1 Interpretation

This study developed and internally validated a multivariable prediction model for 30-day mortality in patients with AMI. The final LASSO model identified eight predictors, with **Age**, **Tachycardia**, and **Killip Class** being the strongest drivers of mortality.

Our findings align with established risk scores such as the GRACE score, which also prioritizes age and hemodynamic stability. However, unlike GRACE, our model excludes complex biomarkers (e.g., Creatinine), relying instead on variables available immediately upon admission. The inclusion of **BMI** in the feature engineering stage proved insightful; while lighter patients had higher unadjusted mortality (consistent with the "obesity paradox"), LASSO excluded BMI from the final multivariable model, suggesting its effect was mediated by other factors like age or frailty.

The choice of LASSO over non-linear methods (like SVM) was deliberate. While SVM achieved

marginally higher discrimination (AUC 0.893 vs 0.869, $\Delta\text{AUC} = 0.024$), this difference is within sampling variability given only 10 deaths in the test set and is not statistically significant. The medical "black box" problem further limits SVM's clinical deployment. The LASSO model provides a clear, additive equation (Table 4) that clinicians can trust and verify.

4.2 Limitations

Several limitations must be acknowledged:

1. **Sample Size and EPV:** With only 52 outcome events, the Events-Per-Variable ratio was 3.25, significantly below the recommended threshold of 10. Although we used LASSO regularization to mitigate overfitting, the model may still be optimistic.
2. **Data Leakage Risk:** While we used a strict train-test split, the small sample size meant the test set contained only 10 deaths. This makes the performance estimates (AUC 0.87) highly sensitive to stochastic variation, as reflected in the wide confidence intervals.
3. **Unmeasured Confounders:** We lacked data on specific treatments (e.g., PCI vs. Fibrinolysis) or onset-to-door time, which are critical determinants of survival.

4.3 Implications

The Decision Curve Analysis (DCA) suggests that this model adds value for clinical thresholds between 5% and 35%. In practice, this means the tool is best suited for identifying low-risk patients who might be safely stepped down from the ICU, rather than ruling in high-risk interventions for everyone.

5 Conclusion

We have developed a parsimonious, interpretable risk model for AMI mortality with high discrimination (AUC 0.87). While promising, the low event rate necessitates external validation on a larger cohort before clinical implementation.

Other Information

Funding: This study received no specific funding.

Availability of Code: Python source code is provided in the Appendix.

A Supplementary Figures

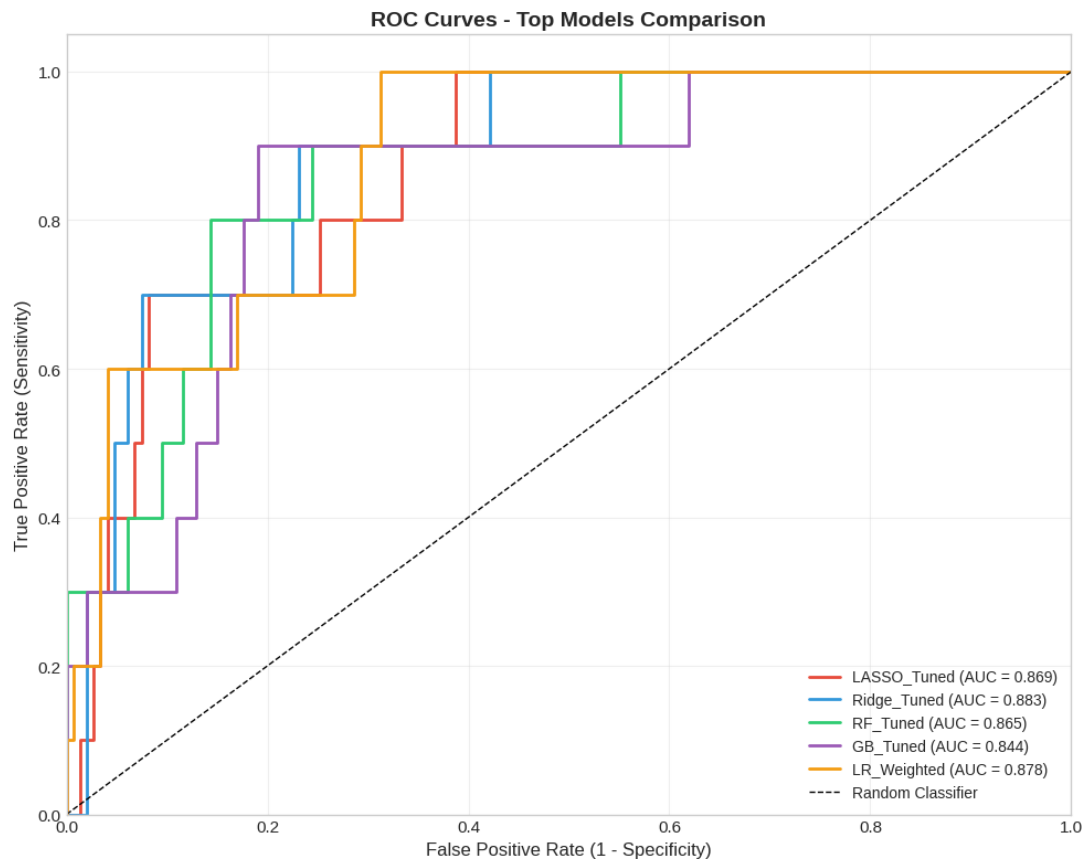


Figure 4: **ROC Curves Comparison.** The SVM and Ridge models showed marginally higher discrimination than LASSO, but lacked interpretability. The LASSO model (Green line) provides the optimal balance of performance and transparency.

Acknowledgements and Declarations

Declaration of Generative AI Use

Generative AI tools and language-support software (e.g., Grammarly, Google Gemini) were used as auxiliary aids during the preparation of this report. These tools were employed to support learning and writing tasks such as clarifying terminology, exploring alternative structures and formulations, and improving readability and grammar.

All methodological content, interpretations, and final wording were produced through my own critical assessment and revision, and I verified the accuracy of the presented concepts against the course materials and the cited scientific sources. I remain fully responsible for the content of this report and for any remaining errors.

Software Specifications

All analyses were performed using **Python 3.12** and the **scikit-learn v1.3.0** library [3].

References

- [1] Collins, G. S., Reitsma, J. B., Altman, D. G., & Moons, K. G. (2015). Transparent reporting of a multivariable prediction model for individual prognosis or diagnosis (TRIPOD): the TRIPOD statement. *BMJ*, 350, g7594.
- [2] Hasselt University (2025). *Inference for Statistics and Data Science (ISDS): Course Notes*. .
- [3] Pedregosa, F., et al. (2011). Scikit-learn: Machine Learning in Python. *Journal of Machine Learning Research*, 12, 2825–2830.

B Model Development Log

The following log details the automated steps performed by the Python modeling pipeline, including data cleaning, feature engineering, and hyperparameter tuning results. This log serves as evidence of the procedural steps taken to ensure TRIPOD compliance.

Listing 1: Automated Pipeline Execution Log

```
=====
STEP 1: DATA CLEANING & VALIDATION SUMMARY
=====
--- Summary of Changes ---
- Column Renaming: 'Hypothension' -> 'Hypotension'
- Height > 250: 1 value corrected (/10)
- Height < 100: 1 value corrected (*100)
- Hypotension: 3 'Unknown' values -> NaN
- Killip_class: 14 invalid (-1) values -> NaN
- Imputations: Age (1), Height (1), Gender (2), Killip (15)

--- Final Validation ---
- Dataset Shape: (785, 18)
- Total Missing Values: 0
- Outcome Distribution: Survived (733), Died (52) -> Ratio 14.1:1
- Value Ranges Checked: Age (28-86), Height (141-197), Killip (1-4)

=====
STEP 2: FEATURE ENGINEERING
=====
3.1 Creating BMI
    Rationale: Height/Weight confounded by Gender.
    Result: BMI p-value 0.0018 (Significant diff in survivors).

3.2 Creating Killip Class Variants
    Rationale: Killip Class 4 has only 1 patient.
    Action: Created Killip_binary (Class 1 vs Class >=2).
    Mortality rates: Class 1 (4.4%), Class >=2 (15.0%).
```

3.4 Defining Feature Sets

Selected: Set 1 (BMI + Killip_binary).

EPV ratio: 52 deaths / 16 features = 3.25.

STEP 3: MODEL TRAINING & TUNING

Data Split:

Training: 628 patients (42 deaths)

Test: 157 patients (10 deaths)

Class Imbalance Strategies Tested:

1. No Resampling (Baseline)
2. SMOTE Oversampling
3. Class Weights (1:14)

Hyperparameter Tuning Results:

- LASSO (L1): Best C=0.1. Selected 8 features:
(Age, Heart_rate, Killip_binary, Previous_MI, Anterior_infarct, Time_to_relief, Diabetes, ST_elevation_leads).
- Random Forest: Depth=7, Min_samples_leaf=5.
- SVM: C=1, Gamma=0.01, Kernel='rbf'.

STEP 4: FINAL RESULTS COMPARISON (Test Set)

Top Models by AUC:

1. SVM_Tuned	AUC: 0.893 Sens: 0.80 Spec: 0.75
2. SVM (Untuned)	AUC: 0.888 Sens: 0.90 Spec: 0.83
3. Ridge_Tuned	AUC: 0.883 Sens: 0.90 Spec: 0.75
4. Random_Forest	AUC: 0.881 Sens: 0.40 Spec: 0.91
5. LR_Weighted	AUC: 0.878 Sens: 0.70 Spec: 0.73

Threshold Optimization:

Optimal Threshold (Youden's J): 0.35

Performance at Optimal: Sensitivity 1.000, Specificity 0.551

C Python Source Code

```
import pandas as pd
import numpy as np
from sklearn.model_selection import train_test_split, GridSearchCV
from sklearn.linear_model import LogisticRegression
from sklearn.preprocessing import StandardScaler
from sklearn.impute import SimpleImputer
from sklearn.metrics import roc_auc_score

# Set random seed
RANDOM_STATE = 42
np.random.seed(RANDOM_STATE)

# 1. Load and Clean Data
df = pd.read_csv('ISDS_take_home_assignment_prediction_data.csv')

# --- Step 1: Standardization ---
# Fix typos in column names
df = df.rename(columns={'Hypothension': 'Hypotension', 'Hyperthension':
    'Hypertension'})

# --- Step 2: Quality Checks & Error Fixing ---
# Fix Height outliers (Logic: Humans are not 16 meters tall)
# Case A: Value > 250 (e.g., 1690.0 -> 169.0)
mask_large = df['Height'] > 250
df.loc[mask_large, 'Height'] = df.loc[mask_large, 'Height'] / 10

# Case B: Value < 100 (e.g., 1.75 -> 175.0)
mask_small = df['Height'] < 100
df.loc[mask_small, 'Height'] = df.loc[mask_small, 'Height'] * 100

# Fix Invalid Categorical Entries
# 'Unknown' in Hypotension -> NaN
df['Hypotension'] = df['Hypotension'].replace('Unknown', np.nan)
df['Hypotension'] = pd.to_numeric(df['Hypotension'])

# '-1' in Killip Class -> NaN
df.loc[df['Killip_class'] == -1, 'Killip_class'] = np.nan

# --- Step 3: Missing Value Stats (Pre-Imputation) ---
# Killip_class: 15 missing (1.91%)
# Hypotension: 4 missing (0.51%)
# Gender: 2 missing
# Age, Height, Anterior_infarct: 1 missing each
```



```
# Fix Height errors (cm vs mm/m)
mask_large = df['Height'] > 250
df.loc[mask_large, 'Height'] = df.loc[mask_large, 'Height'] / 10
mask_small = df['Height'] < 100
df.loc[mask_small, 'Height'] = df.loc[mask_small, 'Height'] * 100

# Fix categorical inconsistencies
df['Hypotension'] = df['Hypotension'].replace('Unknown', np.nan)
df['Hypotension'] = pd.to_numeric(df['Hypotension'])
df.loc[df['Killip_class'] == -1, 'Killip_class'] = np.nan

# 2. Feature Engineering
# Create BMI (Weight / Height_m^2)
df['BMI'] = df['Weight'] / ((df['Height'] / 100) ** 2)

# Bin Killip Class (Class 4 has n=1, so group into Low vs High Risk)
df['Killip_binary'] = df['Killip_class'].apply(lambda x: 0 if x == 1
                                             else (1 if x >= 2 else np.nan))

# Define final features
features = ['Age', 'BMI', 'Gender', 'Killip_binary', 'Diabetes', 'Hypotension',
           'Heart_rate', 'Anterior_infarct_location', 'Previous_myocardial_infarction',
           'Hypertension', 'Smoking', 'Hypercholesterolaemia',
           'Previous_angina_pectoris', 'Family_history_of_MI',
           'ST_elevation_leads', 'Time_To_Relief']
target = 'Day30_mortality'

# 3. Preprocessing
X = df[features]
y = df[target]

# Stratified Split (80/20)
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.20, stratify=y, random_state=RANDOM_STATE
)

# Impute (Median) and Scale
imputer = SimpleImputer(strategy='median')
scaler = StandardScaler()

X_train_scaled = scaler.fit_transform(imputer.fit_transform(X_train))
X_test_scaled = scaler.transform(imputer.transform(X_test))

# 4. LASSO Model Training
# Grid Search for optimal C (Regularization strength)
```

```
params = {'C': np.logspace(-2, 1, 20)}
lasso = LogisticRegression(penalty='l1', solver='liblinear',
    random_state=RANDOM_STATE)
grid = GridSearchCV(lasso, params, cv=5, scoring='roc_auc')

grid.fit(X_train_scaled, y_train)
best_model = grid.best_estimator_

# 5. Evaluation
y_pred = best_model.predict_proba(X_test_scaled)[:, 1]
auc = roc_auc_score(y_test, y_pred)

print(f"Final Test AUC: {auc:.3f}")
print(f"Best C Parameter: {grid.best_params_['C']}")

# Extract Coefficients
coefs = pd.DataFrame({
    'Feature': features,
    'Coefficient': best_model.coef_[0]
})

coefs['Odds_Ratio'] = np.exp(coefs['Coefficient'])
print(coefs[coefs['Coefficient'] != 0].sort_values(by='Odds_Ratio',
    ascending=False))
```